

Integrating district heating potentials into European energy system modelling: An assessment of cost advantages of renewable and excess heat

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ABSTRACT

This paper takes a novel modelling approach by considering high spatial resolution heat generation potentials for district heating and integrating them into a European energy system model. Subsequently, a modelling analysis of an integrated energy system including district heating, electricity and hydrogen supply for 25 EU Member States and the year 2050 is carried out. In contrast to existing approaches, the modelling approach captures the heterogeneous resource availability in district heating. The results show multivalent district heating networks based on a wide range of renewable and excess heat sources used directly or in combination with large-scale heat pumps. The high spatial resolution of the heat generation potentials allows a detailed cost comparison of different possible future technology mixes in district heating. The paper finds that the use of heat pumps, geothermal energy and industrial excess heat offer slight cost advantages for the energy system as a whole. Geothermal heat can also provide cost advantages for district heating generation.

1. Introduction

Achieving the European Union's (EU) target of climate neutrality by 2050 [1] requires rapid progress in decarbonising the energy sector and in making energy demand and supply more efficient and more closely interlinked. To this end, several studies have modelled different scenarios or pathways for future energy systems and discussed in detail the feasibility of achieving this target (see overview in [2]). The existing literature shows that an integrated, holistic perspective is required, which takes into account several sectors, as discussed within the concept of smart energy systems [3–6].

The key to reaching a climate-neutral energy system is to phase out fossil fuels, rapidly expand renewable electricity generation and electrify the other sectors (e.g. heating, transport) as far as possible [2,7]. Moreover, reducing energy demand and supplying energy more efficiently play an important role in line with the “energy efficiency first” principle (EE1st) [8]. As the heating and cooling sector accounts for

about 50 % of European energy demand [9], energy savings in this sector are vital. Efficiency measures in buildings can reduce fuel demand, running costs and investments in (new) heating systems [10]. In the case of district heating, heat savings in buildings allow operators to reduce the temperature level of their grid, making it easier to integrate (low temperature) renewables and excess heat sources (i.e. 4th generation district heating as defined by [11], studied by [12–14] or 5th generation district heating as described by [15]). Thereby, the compatibility of district heating with building refurbishment as a competing or synergetic strategy is still being explored [16]. Even if heat savings in buildings are achieved, the remaining district heating demand still needs to be climate-neutral and efficient. As mentioned above, electrification via heat pumps appears to be a promising option (cf. overview in [2] or project reports [17–19]).

However, renewables such as the direct use of geothermal and solar thermal energy, as well as waste or excess heat from industrial processes, can also make a significant contribution to a climate-neutral and

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efficient district heating supply. District heating networks are very heterogeneous also because renewable and excess heat potentials are unevenly distributed [20,21]. Given the difficulty of representing these local differences, district heating is usually modelled in a spatially simplified way in integrated energy system models covering Europe. There are several projects, such as Heat Roadmap Europe, Hotmaps, ReUseHeat, ReInvest or sEEnergie that have made substantial contributions to energy modelling and provided detailed results for future district heating demand and generation. Moreover, there are scientific publications analysing district heating as part of an integrated energy system (e.g. using the model BALMOREL [22,23] or ENERGYPLAN [24,25]). To the best of our knowledge, the existing European energy system analyses model district heating based on different temperature levels, network sizes or technologies. In contrast to the existing approaches, we use a differentiation of district heating networks based on available renewable and excess heat potentials.

Hence, this paper contributes to the existing literature by exploring a modelling approach for district heating technology mixes based on high spatial resolution generation potentials. Our novel approach integrates differences in district heating generation potentials between and within EU countries into the existing high temporal resolution energy system model ENERTILE (i.e. optimisation of investment and dispatch for district heating, electricity and hydrogen for all hours of the year). The new modelling approach allows a detailed analysis of the impact of district heating technology mixes on system and district heating costs. On this basis, we aim to answer the following research question: Can a high direct use of geothermal, solar thermal and industrial excess heat provide cost advantages for district heating and the overall European energy system in 2050 compared to a high use of heat pumps based on air and water sources?

To address this question, 10 scenarios are modelled for the EU and the year 2050. The analysis builds on preliminary work by Manz et al. [21], which provides the high spatial resolution renewable and excess heat potentials for district heating. A model extension integrates these potentials into the optimisation problem of ENERTILE. Subsequently, each modelled scenario sets a different rate of potential utilisation to account for differences in the exploitation of renewable and excess heat potentials. Two scenario groups are analysed: focusing on (1) high and (2) low temperatures in district heating. Lowering the temperature in district heating networks recognises that improvements in energy efficiency can make a significant contribution to achieving climate neutrality (i.e. EE1st).

The paper is structured as follows: Section 2 describes the methodology and data. Section 3 presents the main results, i.e. the modelling output for district heating and electricity generation as well as the overall system costs and costs for district heating generation. Section 4 discusses the key results, and section 5 draws conclusions.

2. Methodology and data

2.1. Overview of the modelling approach

Regionally heterogeneous resource availability substantially influences the prospects for different technologies in district heating. For example, the availability of geothermal energy depends on the geological situation [21] and industrial excess heat is only available in some areas [26,27]. This means that an approach that assesses the contribution of renewable and excess heat technologies based on high spatial resolution data is necessary. Besides a high level of detail with regard to district heating, interaction with the remaining energy system, e.g. through large-scale heat pumps, also needs to be considered. Therefore,

this paper takes a novel approach in order to model the technology mix in district heating based on high spatial resolution heat potentials in an integrated energy system model.

The analysis uses heat potentials for district heating generation from [21], aggregated into four so-called district heating types. The heat potentials in [21] are calculated using a spatial matching algorithm that maps the heat demand in a given district heating area to the heat generation potentials of different sources. The district heating areas with different heat potentials are then clustered into four district heating types. The district heating types represent multivalent district heating networks, in which specific resources are abundantly available [21]. For example, in *DH type 2: Geothermal*, a significant part of the demand could be covered by geothermal energy due to its favourable geological situation.

The types are integrated into the existing energy system model ENERTILE as separate regions. Subsequently, we carry out a scenario-based analysis for the year 2050 using a green field approach. The green field approach is a solution to overcome data limitations, as data on existing district heating networks and the capacity of different heat plants is lacking in most European countries. Investments in new generation capacity is required in any case by 2050 in order to phase out fossil fuels, as they still represented 59 % of the district heating supply [20]. Existing climate-neutral heat generation plants will also need to invest in maintenance, further arguing in favour of a green field approach.

Fig. 1 provides an overview of the modelling approach. Figure A1 in the appendix visualises the modelling chain and the connection with the preliminary work of [21]. Section 2.2 provides a general overview of the model ENERTILE and section 2.3 describes the model extension with the district heating types.

2.2. The energy system model ENERTILE

ENERTILE is a detailed techno-economic linear optimisation model for energy systems, which minimises the costs of energy supply, transmission and storage (cf. Fig. 1 and [28]). The model has been continuously developed for more than 15 years and has been used in several publications (e.g. [29–32]). A complete documentation of ENERTILE can be found in [33–36].

The model covers the supply of electricity, district heating, and hydrogen. It simultaneously optimises capacity expansion and hourly generation of all system components based on the exogenously specified demand for electricity, district heating and hydrogen. It covers conventional and renewable power plants, combined heat and power (CHP) plants, conventional and renewable heating technologies in district heating networks, cross-border electricity transmission capacities, energy storage technologies, hydrogen generation technologies and infrastructures. Furthermore, the model considers demand-side flexibility from decentralised heat pumps in buildings and electric mobility in the transport sector. The key constraints of the model ensure that the demand for electricity, district heat and hydrogen is met in every region and at every hour of the day. Political decisions and restrictions, such as a CO₂ price and a phase out of fossil fuels, are included to ensure that a climate-neutral energy system is achieved (cf. Fig. 1).

The objective function minimises all costs associated with the vectors of the decision variables for installed capacity $\vec{X}$ and generation $\vec{x}$. For reasons of space and clearness, only an excerpt and simplified version of the objective function of the linear optimisation model is presented (cf. formula 1). Detailed descriptions of ENERTILE and all equations can be found in Refs. [33–36].

$$\min_{\vec{x}, \vec{X}} \left[\text{cost}_{\text{el}}^{\text{fix}}(\vec{X}) + \text{cost}_{\text{el}}^{\text{var}}(\vec{x}) + \text{cost}_{\text{heat}}^{\text{fix}}(\vec{X}) + \text{cost}_{\text{heat}}^{\text{var}}(\vec{x}) + \text{cost}_{\text{el, chp}}^{\text{var}}(\vec{x}) + \text{cost}_{\text{hydrogen}}^{\text{fix}}(\vec{X}) + \text{cost}_{\text{hydrogen}}^{\text{var}}(\vec{x}) \right] \quad (1)$$

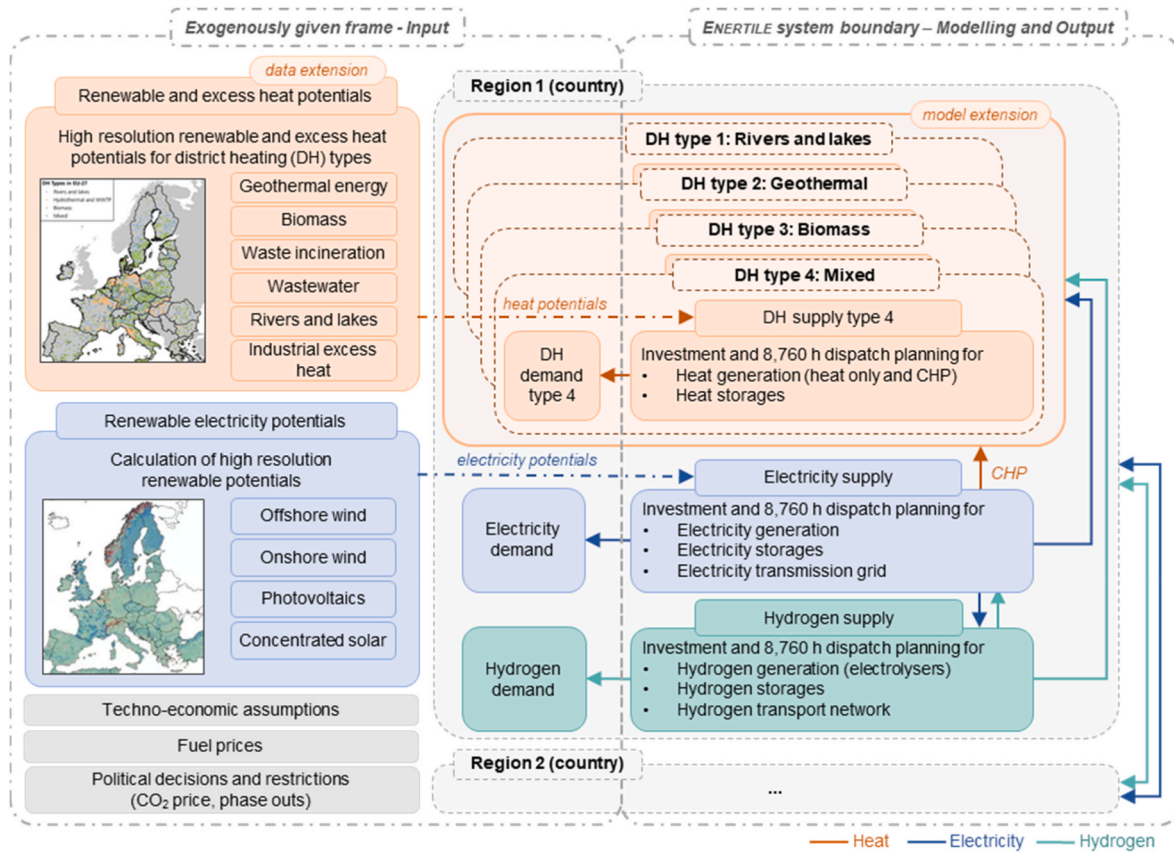


Fig. 1. Schematic representation of the quantities, interactions, and boundary conditions in the model ENERTILE

with

$\text{cost}_{\text{el}}^{\text{fix}}$: fixed costs of electricity capacity expansion in €
 $\text{cost}_{\text{el}}^{\text{var}}$: variable costs of electricity generation in €
 $\text{cost}_{\text{heat}}^{\text{fix}}$: fixed costs of capacity expansion in heat grids in €
 $\text{cost}_{\text{heat}}^{\text{var}}$: variable costs of heat generation in heat grids in €
 $\text{cost}_{\text{el,chp}}^{\text{var}}$: variable costs of electricity generation from CHP plants in heating grids in €
 $\text{cost}_{\text{hydrogen}}^{\text{fix}}$: fixed costs of hydrogen capacity expansion in €
 $\text{cost}_{\text{hydrogen}}^{\text{var}}$: variable costs of hydrogen generation in €

In this paper, ENERTILE's geographical coverage comprises 25 Member States (MS) of the EU.¹ ENERTILE can model several years in a row and has a high temporal resolution with the ability to analyse 8760 h per year. However, in this paper, we use a green field approach and only model the year 2050, as the research question in the paper relates to the cost advantages in the year 2050 (cf. section 1 and 2.1).

The heat supply in district heating and the associated capacity expansion of generation technologies is modelled endogenously in ENERTILE based on the predetermined exogenously heat demand for each region (i.e. district heating types per country). ENERTILE scales the annual demand down to hourly demand, using daily district heating time series that incorporate the average daily outdoor temperatures. Different technologies are available, including electric boilers, large-scale heat pumps, geothermal energy and solar thermal plants, biomass boilers and CHP, waste CHP, hydrogen boilers and CHP, and heat storage. A description of the modelling of the heat generation technologies is

provided in the following section 2.3.

2.3. District heating types and heating technologies in ENERTILE

As stated in section 2.1, data on heat potentials, aggregated into so-called district heating types, is used and integrated into ENERTILE. The four types are based on a clustering analysis where 5815 district heating areas in Europe in 2050 are clustered according to the available climate-neutral heat potentials [21]. Thereby, the cluster analysis has shown that these four district heating types adequately represent the diversity and heterogeneity of the potentials [21].

The types represent multivalent district heating networks which have several sources: (1) ambient heat from rivers and lakes used with heat pumps, (2) ambient heat from wastewater treatment plants used with heat pumps, (3) industrial excess heat used directly,² (4) (deep) geothermal energy (hydrothermal and petrothermal) used directly, (5) biomass and biogas and (6) waste incineration. The district heating types are named based on the heat source with the highest potential, i.e. in *DH type 1: Rivers and lakes*, large-scale heat pumps using rivers or lakes as a heat source could meet about 50 % of the demand. All of the potentials are integrated into the model, except petrothermal potentials due to the low probability of their exploitation, the high costs and the lower specific potentials. Furthermore, solar thermal potentials are not provided by [21], as the spatial availability of technical potentials exceeds the economic potentials. Potentials for other types of heat pumps, e.g. air source heat pumps, are not considered for similar reasons in [21]. Solar thermal energy and air source heat pumps are, however,

¹ Malta and Cyprus do not have a district heating infrastructure and it is assumed that they will not start investing.

² The industrial excess heat provided by Ref. [21] corresponds to climate-neutral heat, i.e. a transformation of the industry is already taken into account.

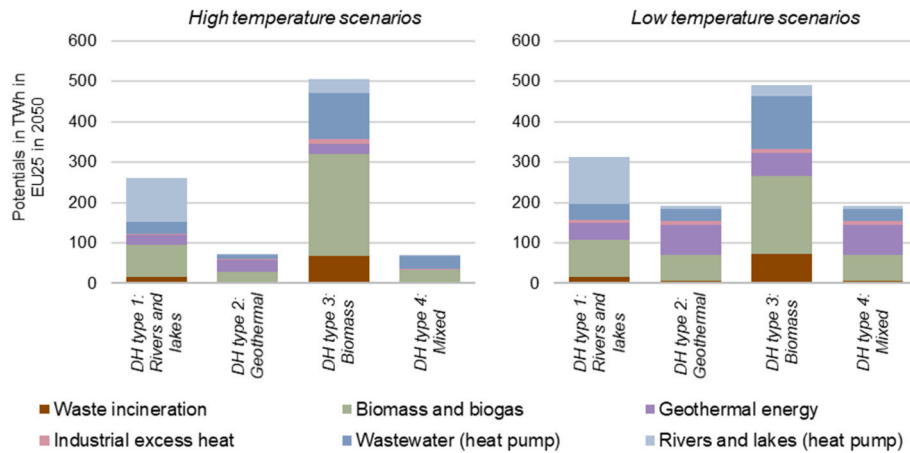


Fig. 2. Potentials per district heating types (own illustration, based on data from [21])⁴

⁴ The difference in potential between the two scenario groups is due to the fact that the potentials are limited to a fixed district heating demand, i.e. 732 TWh in the high temperature scenario and 695 TWh in the low temperature scenario. The district heating demand is lower in the low temperature scenarios because lower heat losses were assumed in Ref. [21].

included in the modelling of this paper as available technologies (cf. below and section 2.4).

Following [21], two different temperature settings for district heating grids are analysed: Higher grid temperatures (flow around 80 °C) and lower grid temperatures (flow around 65 °C).³ With the lower temperatures, a higher share of heat potentials can be utilised. Both temperature settings are used in this paper, leading to two scenario groups: Group (1) includes the potentials for high grid temperatures and group (2) the potentials for low grid temperatures. Fig. 2 illustrates the four district heating types on the European level by showing the aggregated mix of potentials for the two scenario groups.

The four types are integrated into ENERTILE on the MS level, i.e. for each MS, four types of district heating grids are implemented, with the sum of the available renewable and excess heat potentials per type and MS (cf. Fig. 1). As stated, the types represent future district heating areas with the spatially matched heat generation potentials. Thus, the high spatial resolution of the analysis of available potentials is represented with these types (cf. [21]). Even though it is a simplification, this approach makes it possible to represent the available potential for district heating in an aggregated way. Previously, district heating was modelled in ENERTILE with only one grid for each MS. Therefore, the extension of the model strongly improves its representation of district heating.

Furthermore, several technologies have been updated or added to ENERTILE in order to reflect the technological diversity associated with the wide variety of potentials.

(i) Geothermal and solar thermal plants

The use of deep geothermal and solar thermal energy for heat generation in district heating is associated with comparatively high investment and is often also influenced by political preferences. Therefore, the generation of district heating with deep geothermal and solar thermal energy is not subject to the cost optimisation procedure in ENERTILE but is included as an exogenous assumption (cf. [33]). Heat generation from these sources is predefined per district heating type in each MS (in

this paper e.g. based on [21], cf. section 2.4). However, different utilisation rates and the related costs of the two technologies are explored within the scenario design by varying the share of the predefined heat generation from these sources (see section 2.4). Since these renewable heat sources are quasi-inexhaustible, it is assumed that the predefined annual values directly correspond to the heat production in the heat grids. To achieve the required heat production, only the heat flow taken from the water reservoir or the solar collector area and the overall configuration of the system must be adjusted accordingly. Furthermore, a geothermal plant is usually operated as a base load so that constant hourly production is assumed during the year. In contrast, solar thermal generation varies considerably due to the diurnal and seasonal variation of solar radiation. Therefore, an hourly solar irradiance profile is used to derive hourly generation profiles.

(ii) Direct use of industrial excess heat

For this paper, the use of industrial excess heat in district heating is integrated into ENERTILE. The modelling method used for this technology is analogous to geothermal energy. Consequently, the annual heat generation quantities are predefined and converted into a constant hourly generation profile. Again, different utilisation rates and their related costs are explored within the scenario design (cf. section 2.4).

(iii) Large-scale heat pumps

Usually, all large-scale heat pumps in ENERTILE are simplified and modelled as air-source heat pumps. For this paper, the representation of large-scale heat pumps is amplified by integrating additional heat sources like water from rivers and lakes or wastewater. A methodological challenge in modelling heat pumps is how to represent the variable efficiency as realistically as possible. Since the coefficient of performance (COP) of heat pumps strongly depends on the variable temperature of the heat source, the COP is determined in the model in hourly resolution using a piecewise linear approximation function depending on the temperature of the heat source. Based on the assumptions of the inlet and outlet temperatures of the heat source and sink, respectively, and the use of the estimated Lorenz efficiencies of large heat pumps in district heating in Denmark [37], different linear approximation functions for the COP of the heat pumps are derived. Furthermore, two

³ For simplicity, we assume constant flow temperatures in the modelling and have taken the average temperature of each of the two temperature scenarios from Ref. [21].

Table 1
Overview of the modelled scenarios

Scenario name	Utilisation rate of geothermal potential	Utilisation rate of industrial excess heat potential	Contribution of solar thermal as a share of demand
<i>RES</i>	100 %	100 %	10 %
<i>Geo</i>	100 %	50 %	5 %
<i>Solar</i>	50 %	50 %	10 %
<i>IEH</i>	50 %	100 %	5 %
<i>HP</i>	50 %	50 %	5 %

versions of linear functions are used in this analysis to reflect the two different temperature flow settings for district heating grids (high temperature vs. low temperature scenarios). As a result, heat pumps achieve a higher efficiency in the low temperature scenario. For air as the heat source, hourly average temperatures per country are used. Similarly, hourly average water temperatures per country based on [21] are used for rivers and lakes. Furthermore, it is assumed that rivers and lakes can only be used at a water temperature of 3 °C and higher. For wastewater heat pumps, a constant COP is assumed to simplify the calculation. This assumption can be reasoned by the fact that the wastewater temperature is relatively stable and remains around 10 °C even in winter. To incorporate the upper limit of potentials for heat pumps with wastewater and rivers or lakes, additional capacity restrictions for these technologies are included in the linear optimisation with ENERTILE.

(ii) Biomass and waste

The use of biomass for district heating is limited according to the derived potential for each district heating type (cf. [21]). Therefore, maximum generation restrictions limit the possible use of biomass for district heating. The same applies to heat from waste incineration.

2.4. Scenario design

The objective of this paper is to analyse whether local renewable and excess heat sources can provide cost advantages for district heating and the overall energy system (cf. section 1). Therefore, an explorative scenario-based approach is chosen which scans a broad solution space and includes different scenarios for Europe for the year 2050. In line, 10 scenarios are designed by varying the share of specific renewable and excess heat sources in district heating. In particular, in each scenario, the utilisation of geothermal energy, industrial excess heat and solar thermal energy is set at different levels using equal restrictions in the optimisation problem. Thus, a certain share of the district heating generation mix is fixed, and only the remaining supply to meet the demand is freely optimised with the remaining climate-neutral technologies (i.e. heat pumps, biomass and hydrogen boilers and CHP, and waste CHP). This approach makes it possible to analyse the impact of different technology shares on the system and district heating costs.

In order to specify the levels of geothermal energy, industrial excess heat and solar thermal energy, the potential data provided for each district heating type given in [21] are used (cf. section 2.3). In the scenarios, the utilisation rates of the technically available geothermal and industrial excess heat potentials vary from 50 % to 100 %. As previously stated, solar thermal potentials are not provided in [21]. However, solar thermal plants can be utilised (almost) everywhere, as solar radiation is high enough in all European countries [38]. Therefore, solar thermal energy is included with a share expressed as coverage of the demand. In line with the literature [38], this share ranges from 5 % to 10 % in the different scenarios. In contrast, the potentials of rivers and lakes, wastewater, biomass and waste incineration provided by [21] are used as upper limits, and the contribution of these technologies is freely optimised.

Table 1 provides an overview of the scenarios. The first scenario *RES* focuses on an overall high use of renewables and excess heat. Thus, the

utilisation rates of geothermal and industrial excess heat potentials are set to 100 %. Solar thermal is set to a maximal contribution of 10 % of the district heating demand. In the second scenario *Geo*, geothermal potentials are set to 100 % utilisation of the technical potentials, while industrial excess heat and solar thermal are reduced to half of their maximal contribution. Similarly, in the scenario *Solar*, a focus lies on solar thermal energy. Thus, solar is set to its maximal contribution of 10 % of the demand, and the other potentials are reduced to 50 %. In line with this, in the *IEH* scenario, industrial excess heat is set to 100 % utilisation of the potentials and the two other sources are reduced to half of their contribution. Finally, the scenario *HP* focuses on a (potentially) high share of large-scale heat pumps. Therefore, geothermal, solar and industrial excess heat are reduced to half of their maximal contribution in order to have a high residual demand that is optimised. This overall scenario design is used for the high temperature (*HT*) scenario group and the low temperature (*LT*) scenario group, resulting in 10 different scenarios. Fig. 3 provides a graphical illustration of the 10 scenarios.

All of the modelled scenarios achieve climate neutrality in 2050. Assumptions in the other sectors, i.e. all sectors except the district heating supply mix, remain constant in the scenarios (e.g. assumptions for transport or industry). This means that effects caused by changes in the district heating supply mix can be identified, quantified and interpreted.

The scenarios all involve fundamental changes in the district heating technology mix, as district heating in Europe is still dominated by fossil fuels (in 2018: 30 % natural gas, 29 % coal and oil, 7 % non-renewable waste, 27 % biomass, 1 % geothermal, 1 % heat pumps, 2 % industrial excess heat and 2 % others according to [20]).

2.5. Data and assumptions

The future district heating demand in 2050 and heat potentials to cover this demand form the basis of the modelling analysis. The district heating demand for 2050 is based on [21] (building on [39]). Thereby, the future district heating areas are calculated based on future heat demand, distribution costs, and an assumed future district heating market share in the countries (cf. [39]). The resulting district heating demand is kept constant in the scenarios. It represents an increase of current district heating demand to 732 TWh in the *HT* scenarios and 695 TWh in the *LT* scenarios. Compared to current levels of around 445 TWh [20], this means an increase of at least 250 TWh (+50 %).⁵ This ambitious deployment of district heating is overall in line with other energy system scenarios that foresee 400 to 800 TWh for district heating in Europe in 2050 (cf. e.g. [2,10,17,40]).

The district heating demand is used together with heat generation potential data, representing the technical renewable and excess heat potentials with high spatial resolution that are spatially matched to the identified district heating areas (provided by [21]). The spatial matching assumes maximum distances between the different heat sources and district heating areas. The results of the spatial matching are then used in a clustering algorithm to define so-called district heating types, which represent all areas with a similar potential pattern (cf. section 2.3). A

⁵ In the input data for district heating demand (based on [21,39]), it is assumed that the annual useful heat demand decreases due to renovation activities from 3129 TWh in 2020 to 1900 TWh in 2050. This significant decrease of the heat demand leads to a decreased economic competitiveness of district heating, which is compensated with high connection rates between 70 % and 90 %. For the identification of the district heating areas, two conditions are set in Ref. [37]: The annual heat demand supplied in a district heating area must be greater than 5 GWh/a and the average distribution cost must not exceed the pre-defined distribution cost ceiling. Furthermore, the demand projections are based on climate data, i.e. the demand is also influenced by the number of heating degree days. Due to the reduction based on renovation and heating degree days, the increase of the district heating demand is even higher than 50 %.

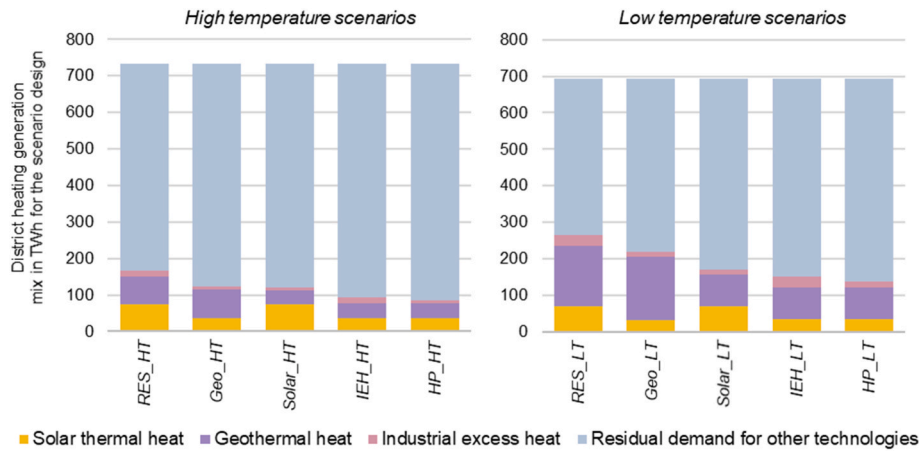


Fig. 3. Overview of renewable and excess heat contributions in the scenarios

detailed description of the methodology, including all assumptions, is given in [21]. The final results are presented and described in [21] and datasets related to [21] can be found in an online data repository.⁶ The geographical distribution of the district heating types in Europe is shown in Figure A2 in the appendix.

In addition to the district heating demand, the scenario runs require demand data for all the other sectors (e.g. electricity demand for appliances in households, electric vehicles or processes in the industry). The primary source for this demand data set is the project ‘Potentials and levels for the electrification of space heating in buildings’ (ENER C1 2019-481) and, thereby, the Elec_60 scenario [18]. This project provides model-based scenarios of the space and water heating sector up to 2050, combining several highly detailed models. In the project, demand data of buildings is provided by the INVERT model, which is then used as input to ENERTILE. Details on the model INVERT can be found in [17,40]. The energy demand is kept constant in all scenarios to ensure that the differences between the scenarios can be compared and interpreted. Table A1 in the appendix presents the input demand data. Losses incurred within energy distribution grids are also accounted for (5.5 % for electricity, and 10 %–16 % for district heating, depending on the temperature level).

Renewable electricity generation is endogenously optimised in ENERTILE (cf. Fig. 1). Thereby, lower limits are defined based on the current electricity generation in the countries [41]. In addition, upper limits for wind and solar are defined on the basis of the calculated generation potential (see methodology in [28,29]). For electricity generation from hydro and geothermal energy, it is assumed that their current production remains constant until 2050 [41].

Lastly, the optimal investment and operation of different technologies (in heating and electricity) depend strongly on the assumed energy carrier prices and techno-economic parameters of the technologies. In all scenarios, a very high CO₂ price of 500 €/t in 2050 is assumed in order to achieve a climate-neutral energy system and to be in line with the input data [18]. Furthermore, an interest rate of 2 % is assumed for all technologies in all scenarios. The techno-economic parameters of the technologies comprise specific investments, variable and fixed operating and maintenance costs, efficiency, lifetime, etc., and are presented in Table A2 in the appendix.

3. Results

3.1. District heating generation

The generation of district heating to meet the given demand is one of the main outputs of ENERTILE. In line with the input demand data, the district heating generation is, in general, higher in the HT scenarios than in the LT scenarios (cf. demand data in section 2.5). Furthermore, the scenario groups and the scenarios have different technology mixes. Fig. 4 shows the modelled district heating technology mixes at the European level in 2050 for the different scenarios.

The technology mixes in the different scenarios show that heat pumps (air, rivers and lakes, wastewater) provide the majority of the heat in all the scenarios. In line with the scenario design, the highest contribution of heat pumps occurs in the HP_HT scenario (459 TWh) and the HP_LT scenario (468 TWh). In terms of capacities, this relates to 209 GW in the HP_HT scenario and 219 GW of capacities for heat pumps in the HP_LT scenario. The lowest contribution of heat pumps occurs in the RES_HT scenario (420 TWh) and the RES_LT scenario (399 TWh).

The results in Fig. 4 show further that river and lake heat pumps and wastewater heat pumps significantly contribute to the share provided by heat pumps overall (9 %–15 % each). Both technologies are optimised with an upper limit according to their potential (cf. section 2). The results show that river and lake heat pumps are on average utilised to around half of their limit, i.e. almost 60 % of the potential is exploited. However, in several district heating types, the river and lake potentials are exploited to 100 %. Wastewater heat pumps are on average utilised to their upper limit, i.e. around 95 %–99 % of their potential is exploited.

As with scenario design, geothermal heat generation is highest in the Geo and RES scenarios. Similarly, solar thermal and industrial excess heat enter the mix in line with the assumptions and the scenario design (cf. section 2.4). In contrast, biomass is optimised in the scenarios with an upper limit. However, biomass shows only slight variations in the scenario groups. Overall, the biomass results show that less biomass is needed in the LT scenarios due to higher potentials in general and higher heat pump efficiencies. Furthermore, in both scenario groups, the biomass shares are slightly higher in the HP scenarios compared to the other scenarios, as less renewable and excess heat potentials are available.

The heat from waste incineration is more or less constant in the scenario groups, with around 6 %–7 %. Electric heaters account for only a very low share of the generation mix in both scenario groups, mainly because of their lower efficiency compared to heat pumps. Lastly, hydrogen-based heat generation technologies contribute only a small amount, around 2 % in the HT scenarios and 1 % in the LT scenarios. Hydrogen-based technologies show low generation but higher capacities. They have a backup role for district heating in times of electricity

⁶ Fordatis - Research Data Repository of Fraunhofer-Gesellschaft; <https://doi.org/10.24406/fordatis/280.2>.

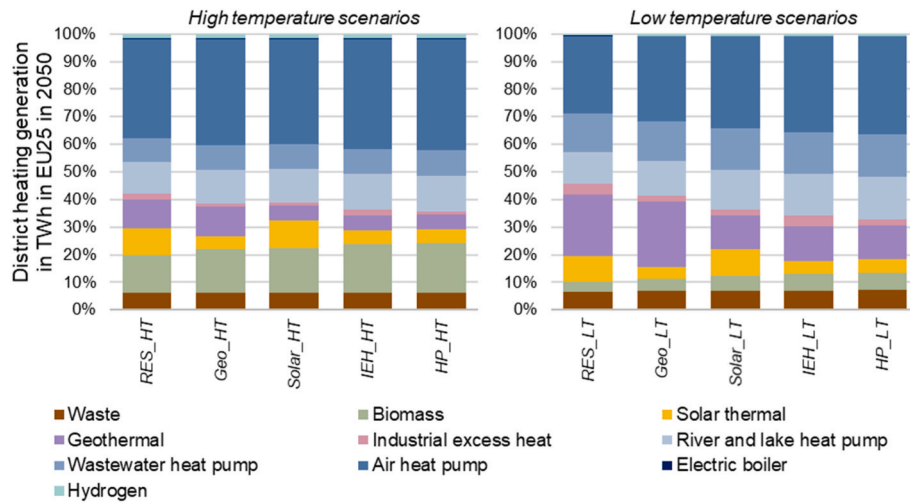


Fig. 4. District heating generation mix in the scenarios in Europe in 2050

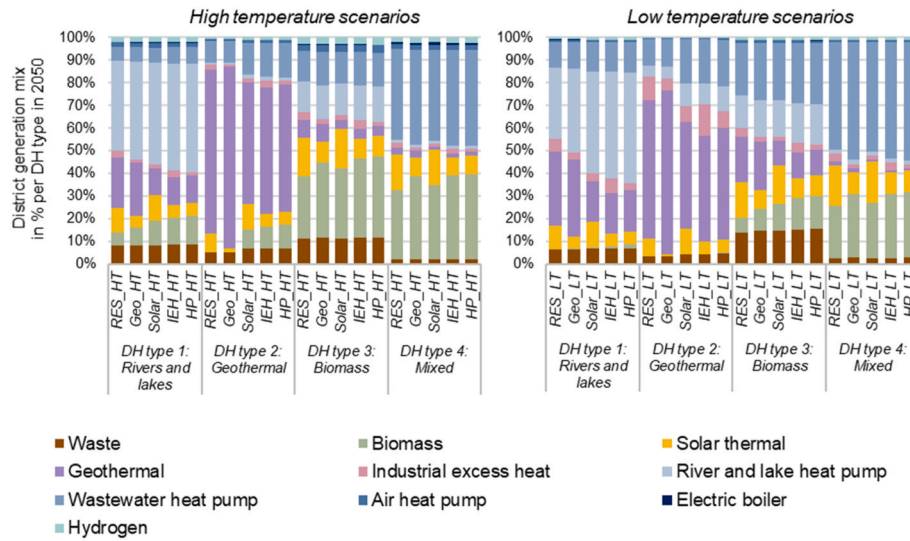


Fig. 5. District heating generation mix in the scenarios per DH type in 2050

shortages as well as shortages of renewable and excess heat.

Fig. 5 shows the district heating generation mix in 2050 in the scenarios per district heating type. In line with the definition of the types, Fig. 5 displays multivalent district heating networks where one or two sources cover a large part of the demand.

The technology mixes in Fig. 5 show that in the HT scenarios, in DH type 1, river and lake heat pumps provide up to 50 % (48 TWh) of the generation. DH type 2 shows a very high share of geothermal energy, up to 80 % (30 TWh). In DH type 3, up to 36 % (54 TWh) is covered by biomass. Finally, DH type 4 shows a generation profile with up to 38 % (10 TWh) biomass and up to 42 % (11 TWh) wastewater heat pumps. Similarly, in the LT scenarios, in DH type 1, river and lake heat pumps can provide up to 49 % (55 TWh) of the generation. In line, DH type 2 shows a very high share of up to 72 % (75 TWh) of geothermal energy. In DH type 3, around 15 % (36 TWh) is covered by biomass. Again, DH type 4 shows a generation profile with up to 29 % (6 TWh) biomass and up to 52 % (11 TWh) wastewater heat pumps. Notably, there are district heating types without the need for biomass, because other sources are available. In particular, in the LT scenarios, DH type 1 and DH type 2 have (almost) no biomass.

Overall, the results for the district heating types underline the diversity and heterogeneity of the potentials, as all types are multivalent

networks using several potentials. The resolution of district heating generation is greatly increased and the heterogeneous resource availability affecting the technology mix in district heating is more adequately reflected by the ENERTILE model extension.

3.2. Electricity generation

This section focuses on electricity generation to assess further differences between the scenarios. Fig. 6 shows the modelled electricity generation mix in the scenarios in Europe in 2050.

One might assume that a higher share of heat pumps in district heating would lead to higher electricity generation, however, the differences between the scenarios are negligible. Overall, in the HT scenarios, electricity generation ranges from 4895 TWh in the RES_HT scenario to 4885 TWh in the HP_HT scenario. Thus, the difference is only 0.19 %. Similarly, in the LT scenarios, generation ranges from 4883 TWh in the RES_LT to 4866 TWh in the HP_LT scenario, i.e. 0.35 % difference.

The maximal generation difference of heat pumps reaches 70 TWh heat (difference between HP_LT and RES_LT). This relates to a 20 TWh difference in electricity needs, which illustrates why the differences in electricity generation are not visible (i.e. very low electricity demand for district heating compared to other sectors).

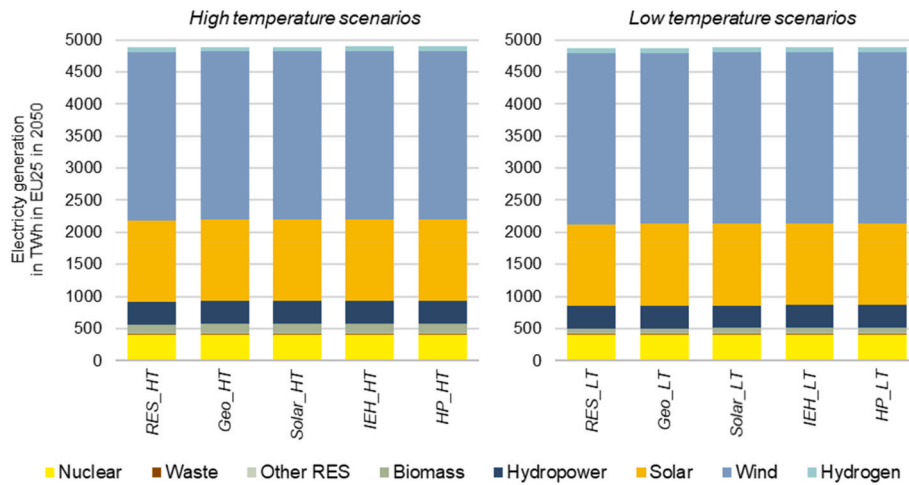


Fig. 6. Electricity generation mix in the scenarios in Europe in 2050

Also, regarding the technology mix, there are no relevant differences. In all the scenarios, electricity generation is greatly dominated by wind and solar power. The need for flexibility in order to balance the energy system increases because of the high share of fluctuating renewable electricity. Energy storage, international trading and demand-side flexibility, electric mobility as well as hydrogen production provide the necessary flexibility in this electricity system of the future.

3.3. System and district heating costs

This last result section focuses on (i) overall system costs and (ii) district heating costs. The objective is to identify the lowest cost scenario, i.e. to analyse whether direct renewables and industrial excess heat in district heating provide cost advantages. Thereby it is important to emphasise that the costs between the *HT* and *LT* scenarios should not be compared. For the *LT* scenarios, additional investments (e.g. in infrastructure and on the building side) and, therefore, capital costs would be necessary in order to reduce grid temperatures. District heating grid costs (e.g. for pipes) and costs on the building side are not in the scope of the ENERTILE modelling. Hence, a comparison of the costs between the scenario groups is not reasonable.

(i) System costs

The system costs (in billion (bn) €) are an output of ENERTILE. They represent the annualised cost of the energy system, covering all fixed and variable costs of the supply side as a whole. Fig. 7 shows the differences between the system costs in the scenarios in Europe in 2050. Thereby the average of the system costs per scenario group serves as a

baseline for the comparison. Thus, only the differences to this baseline are displayed. For example, the system costs in the *RES_HT* scenario are € 1.22 bn higher than the average costs of all *HT* scenarios. The defined baseline reflects the average performance of the scenarios and therefore provides a clear and fair basis for comparison.

Comparing the system costs of the scenarios per scenario group shows a similar picture. In both groups, the *HP* scenario is the cheapest scenario, closely followed by the *Geo* scenario. Furthermore, in both groups, the *IEH* scenario shows lower system costs than the average of the scenarios. In contrast, the *RES* and the *Solar* scenarios show higher than average costs. Thus, independent of temperature levels, higher shares of heat pumps, geothermal and industrial excess heat in district heating seem to be cost-efficient for the energy system.

The system cost comparison shows, however, that the differences between the scenarios are very small. In the *HT* scenarios, the differences are € 2.0 bn, which relates to only 0.7 % of the total system costs (which are around € 280 bn). Similarly, in the *LT* scenarios, the differences are € 2.2 bn, i.e. 0.8 % of the total system costs. Hence, the district heating generation mix has only a minimal impact on the overall European energy system costs. District heating generation reaches up to 770 TWh in the scenarios, while electricity generation reaches almost 4900 TWh (cf. sections 3.1 and 3.2). The relatively low generation of district heating compared to electricity explains the small differences in system costs.

Moreover, sensitivity calculations show that the system costs are very sensitive to the underlying cost assumptions. At the same time, costs for renewable and industrial excess heat projects are still little explored. Geothermal investment, in particular, is highly uncertain. In the scenarios, the investment for geothermal is set to 1300 €/kW [37]

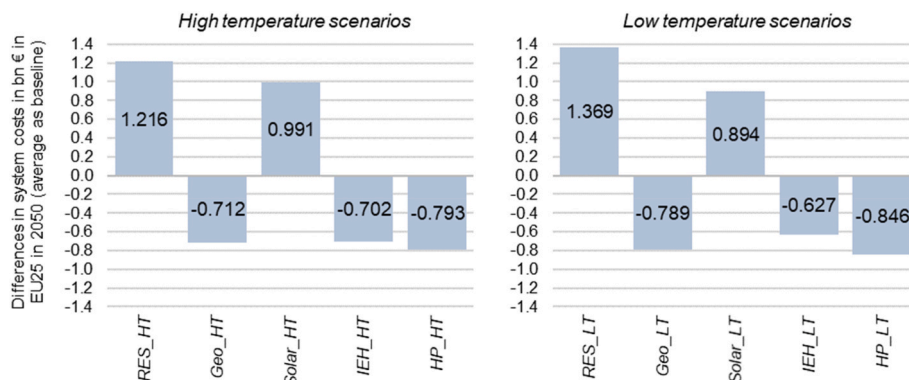


Fig. 7. Difference in system costs between the scenarios in Europe in 2050 (average of the system costs per scenario group serves as a baseline for the comparison)

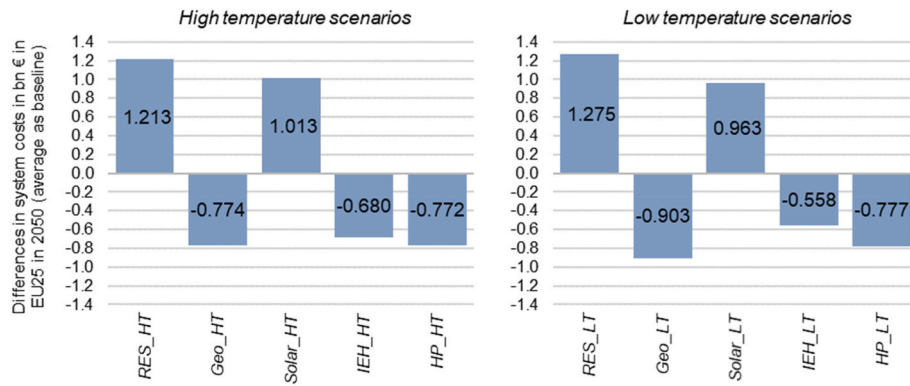


Fig. 8. Difference in system costs between scenarios in Europe in 2050 with 15 % reduced investment in geothermal energy (sensitivity calculation; average of the system costs per scenario group serves as a baseline for the comparison)

(cf. Table A2 in the appendix). If the investment could be reduced by more than 15 %, the overall picture would change, and the *Geo* scenarios would become the cheapest scenarios as shown in Fig. 8.

(i) District heating generation costs

The district heating costs (in €/MWh) represent the cost of district heating generation (in €) divided by the district heating generation (in MWh). Thus, these costs reflect the fixed and variable costs of the total district heating generation (without pipes, substations etc.). Fig. 9 shows the differences between the district heating generation costs in the scenarios in Europe in 2050. Again, the average of the costs per scenario group serves as a baseline for the comparison. Thus, only the differences to this baseline are displayed.

The comparison of the district heating costs between the different scenarios shows that in both scenario groups, the *RES* and the *Geo* scenarios are below average, while the other scenarios are above average. Thus, high shares of renewable heat, in particular geothermal heat, seem to offer cost advantages for district heating. Overall, the district heating generation mix has a greater impact on district heating generation costs than on total energy system costs. The differences range from up to 2.66 €/MWh (4 % of total district heating generation costs of 66 €/MWh) in the *HT* scenarios and even up to 4.62 €/MWh (7 %) in the *LT* scenarios. Particularly for low-income households, which already spend a high proportion of their income on energy, a cost difference of 4.62 €/MWh is a significant amount.

4. Discussion

The optimisation results show that in all the scenarios heat pumps achieve a high share in the district heating generation mix in Europe in 2050. Even in the scenarios focusing more on direct renewable and industrial excess heat, heat pumps cover more than half of the demand. The main reason for this is the cost efficiency of heat pumps compared to alternative technologies (i.e. biomass or hydrogen boilers and CHP etc.). In addition to their economic advantage, our results show that heat pumps also provide valuable flexibility to the electricity system (in line with a more detailed analysis of the flexibility of heat pumps in [29]).

The high share which heat pumps account for is in line with the results from other scenario analyses [2,17,18,42]. Compared to these existing studies, our results provide more technological detail and a more differentiated view, showing that river and lake heat pumps and wastewater heat pumps can make a significant contribution to the district heating generation mix in 2050 (up to 15 % each). These heat pumps are more efficient than air-source heat pumps, and their deployment should be the focus of further research. Furthermore, additional sources for heat pumps (e.g. near-surface geothermal energy and excess heat from data centres, shopping malls, metro stations and electrolyzers) should be explored in future research.

According to the results, other generation technologies also have relevant contributions to a climate-neutral and efficient district heating generation mix. Geothermal, solar thermal, and industrial excess heat enter the mix in line with the scenario design and assumptions. Biomass

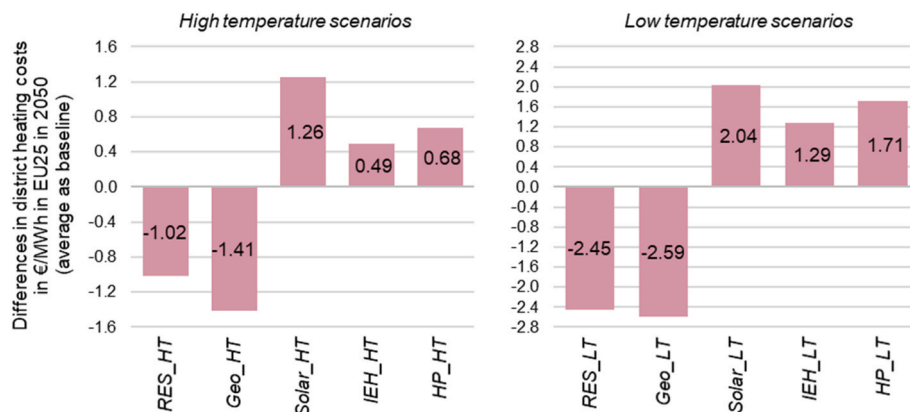


Fig. 9. Difference in the district heating costs between the scenarios in Europe in 2050 (average of the costs per scenario group serves as a baseline for the comparison)

shows a higher generation share in the high temperature (*HT*) scenarios than in the low temperature (*LT*) scenarios. This is mainly because of lower geothermal and industrial excess heat potentials and lower efficiencies of heat pumps in the *HT* scenarios. However, sensitivity analyses show that biomass heat generation is very sensitive to the assumed price. With a higher price (i.e. more than 13 €/MWh), even less biomass would be used. An existing study also demonstrates that district heating can reduce biomass dependency more than individual solutions [43]. Overall, the role of (sustainable) biomass and the various uncertainties regarding biomass prices should be analysed in future research. Regarding the power sector, the results show only negligible variation in electricity generation between the scenarios. The electricity demand increase in scenarios with a higher share of heat pumps is less than 1 %. This is mainly attributed to the high efficiency of heat pumps.

Similarly, the differences in the overall system costs between the scenarios are very low with less than 0.8 %. Thus, the developments in the district heating mix seem to have only a very limited impact on the overall system costs. Nevertheless, the results show that higher shares of heat pumps, geothermal and industrial excess heat in district heating are cost-efficient for the energy system. In particular, the scenarios with the higher shares of heat pumps achieve the lowest system costs. This can be explained by their comparatively low (levelised) costs of heating and by their flexible contribution to the power system, helping to integrate a high share of fluctuating renewable electricity.

However, our results and in particular the system cost comparison is very sensitive to the techno-economic assumptions. Thereby it should be noted that in the modelled scenarios, relatively high, conservative cost assumptions are made for geothermal energy. With 15 % lower investment costs for geothermal plants, the scenario with higher geothermal contribution could become the most cost-efficient one. Higher shares of geothermal energy would lead to accordingly lower shares of heat pumps. Hence, further research should focus on the costs of direct renewables and industrial excess heat and their impact on the district heating mix and the energy system. Analyses with multiple cost sensitivities that comprehensively cover uncertainty ranges can thereby build on the methodological approach and initial findings of this paper.

In contrast, there are greater differences in district heating costs between the scenarios reaching 4 % in the *HT* and 7 % in the *LT* scenarios. Thereby, higher shares of geothermal energy seem to offer cost advantages for district heating. Higher shares of solar thermal energy cannot provide cost advantages for district heating according to our modelling results. This is most likely reasoned by the fact that the profile of solar thermal energy is less suited to the demand curve. However, it is important to note that the modelling analysis does not include seasonal storage,⁷ which could overcome this disadvantage. Further research with seasonal storage should be carried out to analyse the optimal contribution of solar thermal energy optimally with several cost sensitivities.

Overall, our modelling approach uses cost assumptions per technology that do not vary by country or region. In reality, however, investment in the technologies modelled may be more expensive in some regions than in others due to geological and financing conditions or labour costs. Therefore, analyses with further cost sensitivities and possibly regional cost variations to the extent possible are crucial for future research to complement and extend our findings.

In addition, it is important to stress that the results are obtained for a fixed district heating demand in 2050 (732 TWh in the *HT* and 695 TWh in the *LT* scenarios; cf. section 2.5). Different trends in district heating demand can have a large impact on potentials and costs. Hence, further research could explore how different paths of district heating demand would affect the results of the scenarios analysed in this paper. In this context, it would also be interesting to investigate whether different

technology options for district heating supply could change the competitiveness of district heating compared to other heat supply options (e.g. individual heat pumps).

Lastly, it is important to emphasise that the results are highly dependent on the input data and assumptions and are subject to a wide range of uncertainties. In order to reflect these uncertainties, further analyses are needed that cover both the whole system perspective to capture cross-sectoral and international dependencies, and regional analyses that can take into account more granularity and variation. Our results show that, in particular, large-scale heat pumps are cost-effective for the energy system and geothermal energy offers cost advantages for district heating. However, as the cost differences are relatively small, no climate-neutral technology should be excluded in future research and especially in practical implementation projects of district heating networks. For each district heating network, all possible potentials should be considered and their suitability analysed from different perspectives at the local level.

5. Conclusions

This paper takes a novel approach and models different technology mixes in district heating for 2050 in Europe taking into account interaction with and the impacts on the power sector while simultaneously using high spatial resolution renewable and excess heat potentials for district heating generation. For this, the analysis uses heat potentials, aggregated into four district heating types, and integrates them into the existing energy system model ENERTILE. Consequently, the representation of district heating in ENERTILE is significantly improved.

A set of 10 scenarios is used to analyse different rates of potential exploitation and the impact on the overall system and district heating costs. In the process, one set of scenarios focuses on high temperatures and one on low temperatures in district heating networks. The regionally heterogeneous resource availability which affects the technology mix in district heating is more accurately reflected by this novel modelling approach. Nevertheless, future research is needed on the required resolution of district heating in integrated energy system models.

The optimisation results show that at the European level, heat pumps achieve high shares in the district heating generation mix in all the scenarios. Besides air-source heat pumps, river and lake and wastewater heat pumps can significantly contribute to the future generation mix (with up to 15 % each). The advantages of lower grid temperatures are an increase in heat generation potentials, improved efficiencies for heat pumps and a reduced need for biomass. Nevertheless, (some) disadvantages, such as the need for investment in infrastructure and buildings in order to heat homes at lower temperatures, must be considered. Future research should focus on optimal grid temperature levels, e.g. with different temperature type networks.

With regard to system costs, the results indicate that the developments in the technology mix in district heating networks in the analysed scenarios only have a very limited impact on the total European energy system costs. It might be expected that higher shares of heat pumps in district heating could lead to higher electricity generation and thus higher system costs, but the differences between the scenarios are negligible, mainly due to the high efficiency of heat pumps. Nevertheless, higher shares of heat pumps, geothermal energy and industrial excess heat in district heating are cost-efficient for the overall energy system. Scenarios with the highest share of heat pumps are the cheapest, probably due to low generation costs and their role in providing flexibility in the power sector. However, system costs are very sensitive to the cost assumptions, which makes further research a necessity.

When looking at district heating costs, the results show that a higher share of geothermal energy is cost-efficient for district heating. Hence, the research question of the paper can be answered to the extent that a high direct use of geothermal can offer cost advantages for district heating, but does not have a relevant impact on the costs of the overall

⁷ In the district heating networks, water storage tanks that can store heat for several days are modelled, but no seasonal storages have been included so far.

energy system. Future research should focus on the deployment of renewable and excess heat in district heating to achieve a rapid market uptake and a climate-neutral energy system in 2050.

CRediT authorship contribution statement

Anna Billerbeck: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christiane Bernath:** Writing – original draft, Validation, Software, Methodology. **Pia Manz:** Writing – review & editing, Validation, Resources, Data curation. **Gerda Deac:** Validation, Software. **Anne Held:** Writing – review & editing, Validation. **Jenny Winkler:** Writing – review & editing. **Ali K  k:** Writing – review & editing. **Mario Ragwitz:** Supervision.

Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used “deepl write” to correct potential grammar mistakes and improve readability. After using this tool/service, the authors reviewed and edited the content as

needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The input data are cited with references, listed in the appendix and published at Fordatis - Research Data Repository of the Fraunhofer-Gesellschaft; <https://doi.org/10.24406/fordatis/280>

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Appendix

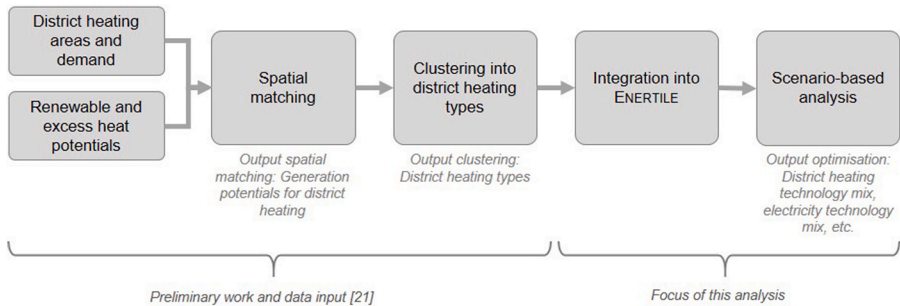


Fig. A1. Overview of the modelling chain and connection to the preliminary work of [21]

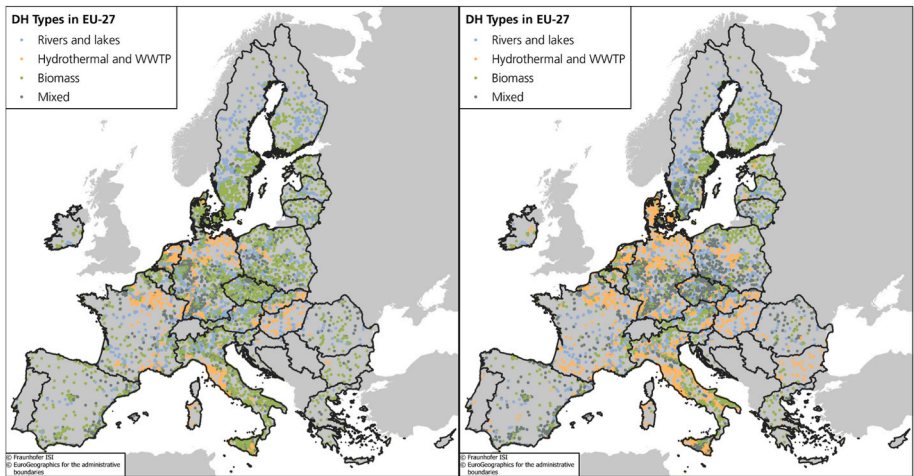


Fig. A2. Location of the DH types in the HT scenario (left) and LT scenario (right) (based on [21])

Table A1

Energy demands per country used as input in ENERTILE in 2050

Country	Demand in TWh				
	District heating demand [21]		Demand other sectors ([18] building on [44,45])		
	HT scenario	LT scenario	Heat demand decentralised heat pumps in buildings	Electricity demand	Hydrogen demand
Austria	18.5	17.5	19.3	89.9	27.0
Belgium	8.6	8.2	26.3	121.8	74.5
Bulgaria	6.4	6.1	6.3	37.7	7.8
Cyprus	0.0	0.0	0.3	5.0	0.2
Czechia	28.8	27.3	27.8	81.5	14.2
Germany	148.9	141.2	120.1	734.4	143.6
Denmark	21.2	20.1	11.2	53.2	0.3
Estonia	4.0	3.8	2.7	9.4	0.5
Spain	44.1	41.8	29.0	308.5	29.2
Finland	34.9	33.1	14.5	98.1	6.6
France	107.3	101.8	149.7	586.3	148.0
Greece	9.4	8.9	4.9	59.1	6.5
Croatia	3.4	3.2	3.6	17.0	0.8
Hungary	17.0	16.1	25.4	52.3	11.8
Ireland	1.3	1.3	11.2	32.8	2.4
Italy	89.3	84.7	74.7	372.3	50.4
Lithuania	6.9	6.6	3.6	13.9	5.5
Luxembourg	2.9	2.7	2.4	9.8	0.0
Latvia	4.9	4.6	3.0	10.0	0.0
Malta	0.0	0.0	0.0	2.5	0.1
Netherlands	24.7	23.4	40.9	172.7	56.7
Poland	59.3	56.3	40.5	223.5	23.4
Portugal	3.6	3.4	5.2	67.2	7.4
Romania	12.5	11.8	14.6	77.4	22.7
Sweden	65.8	62.4	25.1	169.9	11.4
Slovenia	1.7	1.6	19.3	18.0	0.0
Slovakia	6.9	6.5	26.3	39.4	9.2
Total	732	694	708	3463	660

Note: The data publicly available from the 1.5TECH is limited to overall values for Europe in 2050 [44]. Thus, the demand distribution between countries were based on the SET-Nav project (diversification pathway) [45].

Table A2

Techno-economic parameters of investment options in ENERTILE in 2050 [37,46]

Technology		Lifetime	Investment in €/kW	Fixed O&M in €/kW	Variable O&M in €/MWh	Efficiency el (chp)	Efficiency heat
Electricity	Battery storage	10	204	5.5	0.0	95 %	
	Combined cycle hydrogen turbine	30	750	11.3	3.0	61 %	
CHP	Hydrogen turbine	30	400	7.5	1.5	41 %	
	Biomass CHP	25	2900	103	2.1	30 % (71 %)	
	Waste to energy CHP	25	6460	148	23.6	24 % (80 %)	
	Hydrogen CHP	30	950	30	3	48 % (88 %)	
District heating	Biomass boiler	25	750	42.9	0.7		103 %
	Electric boiler	20	60	0.9	0.4		99 %
	Air source heat pump	25	760	2.0	1.7		variable
	River and lake heat pump	25	380	4.0	1.7		variable
	Wastewater heat pump	25	570	2.0	1.7		variable
	Geothermal	30	1300	18.8	2		–
	Solar thermal	30	310	60	0		–
	Industrial excess heat	25	1500	80	5.6		–
	Heat storage	20	22	0.0	0.0		99 %
	Hydrogen boiler	25	50	1.7	0.9		104 %

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